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SUPERSEDING
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MILITARY SPECIFICATION

POWDER, SULFURLESS MEAL

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gunpowder designated as sulfurless meal powder for use in munitions.

1.2 Classification. Sulfurless meal powder shall be furnished in the following grades and classes, as specified (see 6.1):

Grade A - Unglazed.
Grade B - Glazed.
Class 1 - Finely granulated.
Class 2 - Coarse.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard, Testing.

MILITARY

JAN-P-156 - Potassium Nitrate.
JAN-C-178 - Charcoal (For Use in Munitions).

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129 - Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 Other publications. The following publication, of the issue in effect on the date of invitation for bids, unless otherwise stated, forms a part of this specification:

INTERSTATE COMMERCE COMMISSION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the powder shall conform to this specification.

3.2 Chemical. The sulfurless meal powder shall conform to the chemical requirements shown in table I.

Table I - Chemical requirements

Requirement	Maximum	Minimum
Potassium nitrate, class a (Specification JAN-P-156) parts by weight	71.0	70.0
Charcoal (Specifications JAN-C-178) parts by weight	30.0	29.0
Free sulfur (percent)	0.1	----
Moisture (percent)	2.2	0.7

3.3 Particle size. The powder shall be granulated to conform to the particle size requirements shown in table II.

Table II - Particle size requirements

Particle Size	Class 1	Class 2
	Percent	Percent
Through a No. 10 sieve (min.)	----	100
Retained on a No. 18 sieve (min.)	----	75
Through a No. 20 sieve (max.)	----	6
Through a No. 140 sieve (min.)	100	----
Through a No. 230 sieve (min.)	75	----

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3.4 Grade A.

3.4.1 Carbon. The carbon content of the charcoal in the grade A powder, when tested as specified in 4.4.5, shall be between 76.0 and 82.0 percent, calculated on the dry, ash-free basis.

3.4.2 Ash. The charcoal, in the grade A powder, shall yield not more than 3.0 percent ash, when tested as specified in 4.4.5.

3.4.3 Flashing. The grade A powder, when tested as specified in 4.4.9, shall burn with very few sparks, leaving a finely divided residue, weighing not more than 5 percent of the weight of specimen. The residue shall be free from decided specks of fused salt or carbon.

3.4.4 Pelleting. The load required to break the pellet of grade A powder, when prepared and tested as specified in 4.4.10, shall be not less than 600 grams.

3.4.5 Finish. The finished grade A powder shall be unglazed.

3.5 Grade B.

3.5.1 Specific gravity. The specific gravity of the Grade B powder, when tested as specified in 4.4.11, shall be 1.61 $\pm$ 0.03.

3.5.2 Finish. The finished grade B powder, after granulation, shall be glazed by friction without the aid of graphite.

3.6 Workmanship. The powder shall be incorporated thoroughly by an approved process that insures the required uniformity of the finished product. Any proposed change in or accidental deviation from the approved process, however slight, shall be reported to the inspecting officer. No powder produced under such change or deviation shall be offered for acceptance until such change or deviation is approved by the contracting officer.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.

4.1.1 Batch process. A lot shall consist of the material produced from no more than one manufacturing batch (see 6.2).

4.1.2 Continuous process. A lot shall consist of the material produced in no more than one work shift with no change in process or materials.

4.2 Sampling. Five 1-pound specimens shall be selected as representative of each lot. The specimens may be taken randomly from containers or during the container filling operation. The specimens shall be placed in clean, dry containers, and labeled to identify the container

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and lot represented. Each specimen shall be tested according to 4.4 (see 6.3).

4.3 Inspection.

4.3.1 Packing and marking. The inspector shall inspect the packing and marking of the material for compliance with section 5.

4.3.2 Procedures. For Navy purchases, the general inspection procedures shall be in accordance with the General Specifications for Inspection of Material.

4.3.3 Raw materials. The inspector shall ascertain that all raw materials procured under separate specifications have been inspected, tested, and accepted in accordance with their respective specifications or applicable standards.

4.4 Tests.

4.4.1 Reagents. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with each test, using the same quantities of reagents used in the test, and corrections shall be applied when significant.

4.4.2 Potassium nitrate. Weigh, to the nearest milligram (mg.), approximately 10 grams (gm.) of the specimen into a tared fritted glass crucible (see 6.5). Draw through the crucible, by means of suction, about 200 milliliters (ml.) of hot water in successive portions of 10 to 15 ml. each. Test the final portions of water passing through the crucible with an excess of strong sulfuric acid containing a few crystals of diphenylamine until there is no blue color reaction indicating the presence of nitrates. Dry the crucible for 4 hours at 70 ± 1°C. and weigh. Consider the percent loss in weight minus the moisture content (4.4.8) as representing the potassium nitrate. (It is recognized that this included any water-soluble portion of the charcoal or sulfur.) Retain the crucible and residue for determination of sulfur and charcoal.

$$\text{Percent } KNO_3 = \frac{100\ (A - B)}{A} - C$$

where: A = Weight of the specimen, in gm.
B = Weight of the residue, in gm.
C = Percent moisture of the specimen (4.4.8).

4.4.3 Sulfur. Place the crucible, weighed for loss of water-soluble material (see 4.4.2), in an extractor on a water bath, and extract for 1 hour with carbon disulphide. After extraction, wash the crucible once with alcohol and once with ether, using suction, and dry for 1 hour at 105 ± 3°C. Calculate percent sulfur from the loss in weight.

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$$\text{Percent sulfur} = \frac{100\ (B - C)}{A}$$

where: A = Weight of the specimen, in gm.
B = Weight of the unextracted charcoal, in gm.
C = Weight of the carbon disulfide-extracted charcoal, in gm.

4.4.4 Charcoal. Consider the residue in the crucible (4.4.3) to be charcoal. Determine its weight by subtracting the tared weight of the crucible from the weight of the crucible and residue. Calculate as percent charcoal. Retain the residue for determination of carbon and ash.

$$\text{Percent Charcoal} = \frac{100\ C}{A}$$

where: A = Weight of the specimen, in gm.
C = Weight of the charcoal, in gm.

4.4.5 Carbon and ash. Dry the charcoal residue (see 4.4.4) at 140 ± 5°C. in a current of nitrogen (free from oxygen) and cool in a desiccator. Determine the carbon on a suitable portion of the residue by combustion and absorption in the usual manner. Weigh the ash remaining after combustion.

4.4.6 Particle size. Select U.S. Standard sieves conforming to Specification RR-S-366 for the class specified in 3.3. Nest the sieves in order of decreasing size, placing the largest mesh on top and a receiving pan at the bottom. Weigh, to the nearest centigram (cg.) approximately 25 gm. of the material and place it in the top sieve. Brush the material with a camel's-hair brush through the smallest mesh that it will pass and calculate the percent retained on each sieve.

4.4.7 Moisture. Weigh, to the nearest mg., approximately 2 gm. of the specimen into a tared weighing dish or covered watch glass. Dry 2 hours in an oven at 100 ± 2°C. Cover, cool in a desiccator, weigh, and calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100\ (A - B)}{A}$$

where: A = Weight of the specimen, in gm.
B = Weight of the specimen after drying, in gm.

4.4.8 Flashing. Distribute approximately 5 gm. of the specimen, weighed to the nearest cg., evenly over the bottom of a tared, thin, flat-bottom dish, 4 inches in diameter by 3/4 inch in depth, and "flash" the powder. Weigh the residue and examine visually for conformance to 3.4.4.

$$\text{Percent residue} = \frac{100\ A}{B}$$

where: A = Weight of the residue, in gm.
B = Weight of the specimen, in gm.

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4.4.9 Pelleting. Thoroughly dry approximately 10 gm. of the powder in an oven at 70 ± 1°C to a constant weight and then condition to a moisture content of 1.75 ± 0.05 percent. By means of a suitable mold, compress 0.4 gm. of the conditioned powder, weighed to the nearest mg., into a disk 0.50 inch in diameter under a dead load pressure of 1,000 pounds, maintained for 15 seconds. Remove the pellet from the mold and center over a cavity of 0.40-inch diameter. By means of a metal rod, 0.085-inch thick, cut perpendicular to the line of axis at the end, apply pressure to the center of the pellet and note the load required to break the pellet. The average of three tests shall conform to 3.4.5.

4.4.10 Specific gravity. Transfer, approximately 10 gm. of specimen, weighed to the nearest mg., to a 50-ml. specific gravity bottle which has been one-half to two-thirds filled with clean mercury. Displace the remaining air in the bottle with mercury as follows: Attach a piece of heavy rubber tubing approximately 2 feet long to the bottle. Connect the other end of the tubing to one arm of a Y-tube. By means of another piece of rubber tubing and a short piece of glass tubing, extend the other arm of the Y-tube to dip into a vessel containing mercury. Connect the stem of the Y-tube with a vacuum line. Close the rubber tubing leading to the mercury vessel with a pinchcock. Open the vacuum line in order to evacuate the specific gravity bottle. After evacuation, close the vacuum line with a pinchcock placed close to the Y-tube. Open the pinchcock on the tubing leading to the stem of the mercury vessel and allow the mercury to flow into the bottle. Repeat this operation until the bottle is filled with mercury. Weigh the bottle containing the powder and mercury at 20 ± 0.3°C. Weigh the same bottle filled with mercury at 20 ± 0.3°C. Calculate the specific gravity of the powder at 20°/20°C. as follows:

$$\text{Specific gravity at } 20°/20°C. = \frac{13.55 \times A}{A + B - C}$$

where: A = Weight of specimen, in gm.
B = Weight of bottle filled with mercury, in gm.
C = Weight of bottle filled with mercury and specimen, in gm.

4.4.11 Finish. Visually inspect the material for compliance with 3.5.2.

4.5 Rejection and resubmission. If any specimen fails to conform to this specification, the lot shall be rejected. The contractor, at no expense to the Government, shall have the option of having an analysis made on each container in the lot, removing the nonconforming material, and resubmitting the remaining portion of the lot for acceptance testing. If any speciment taken from the resubmitted portion of the lot fails to conform to this specification, the lot shall be finally rejected.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. 25 ± 1 pounds of the material shall be packaged in a 4-ounce cotton duck bag; 1-1/2 pounds of the material shall be packaged in metal kegs conforming to Specification 13 of the Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc. or in fiber or metal containers; or the material shall be packaged in cylindrical fiber cartridges not over 5 inches in diameter nor over 18 inches in length with the fiber paraffined at least 0.05 inches on the outside and with all joints securely glued or cemented.

5.2 Packing. The cotton bags, metal kegs, or fiber or metal containers shall be packed in wooden boxes, conforming to Specification 14 of the Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., having a gross weight not exceeding 140 pounds or shall be packed in wooden boxes, conforming to Specification 15A or 16A of the Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., having a gross weight not exceeding 200 pounds. The fiber cartridges shall be packed in wooden boxes, conforming to Specification 14, 15A, or 16A of the Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., having a gross weight not exceeding 75 pounds or shall be packed in fiberboard boxes, conforming to Specification 23F of the Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., lined with a strong paraffined paper or other suitable waterproofed material without joints or other openings at the bottom or sides and having a gross weight not exceeding 65 pounds.

5.3 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., and Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Grade and class of material required (see 1.2, 3.3, 3.4, and 3.5).
c. Type of packaging required (see 5.1).
d. Type of packing required (see 5.2).

6.2 Batch. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

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6.3 Sampling and testing. When the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.4 Safety classification. This item carries Ordnance safety classification 9, storage group none.

6.5 Crucible. A fritted glass Gooch-type crucible, such as described in the Central Scientific Company's Catalog J-150, under stock number 13131 is satisfactory for the analysis in 4.4.2.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor nay obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-12059 Powder, Sulfurless Meal

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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